Zihan Yang

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About Me

I am current a research student at Beihang University with a background in materials science and aerospace engineering, now focusing on robotics. My research interests lie at the intersection of control and machine learning, covering both learning-based planning and control, as well as control-theoretic methods that improve learning algorithms. Recent works have been recognized at ICRA 2026, ICLR 2025 (Oral) and ICCA 2024 (Best Student Paper Award).

Education

Northwestern Polytechnical University, B.Eng. in Material Science and Engineering — Sept 2019 - June 2023

Queen Mary University of London B.Eng. in Material Science and Engineering (*UK-China Transnational Education*) — Sept 2019 - June 2023

- GPA: 3.59/4
- **First Class Honors.**

Beihang University, M.S. in Aerospace Science and Technology — Sept 2023 - Jan 2026

- GPA: 3.80/4, supervised by Prof. Kexin Guo, Prof. Xiang Yu, and Prof. Lei Guo

Publications

(* equal contribution)

[1] **Z. Yang**, J. Jia, M. Wang, Y. Liu, K. Guo, and X. Yu, "Unified meta-representation and feedback calibration for general disturbance estimation," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[2] J. Jia*, M. Wang*, **Z. Yang**, B. Yang, Y. Liu, K. Guo, and X. Yu, "Learning-based observer for coupled disturbance," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[3] J. Jia*, **Z. Yang***, M. Wang, K. Guo, J. Yang, X. Yu, and L. Guo, "Feedback favors the generalization of neural ODEs," in *The Thirteenth International Conference on Learning Representations (ICLR)*, **Oral Presentation (acceptance rate: 1.8%)**, 2025.

[4] K. Guo*, **Z. Yang***, J. Jia, Y. Liu, and X. Yu, "Optimizing control-friendly trajectories with self-supervised residual learning," *IEEE/ASME Transactions on Mechatronics (Under Review)*, 2025.

[5] **Z. Yang**, J. Jia, Y. Liu, K. Guo, X. Yu, and L. Guo, "TRACE: Trajectory refinement with control error enables safe and accurate maneuvers," in *IEEE 18th International Conference on Control and Automation (ICCA)*, **Best Student Paper Award**, 2024.

[6] X. Song, X. Guo, G. Liu, **Z. Yang**, and L. Liu, "Co-optimization of motion and energy domain for hydrogen-powered hybrid UAVs: A bi-directional coupling architecture," in *IEEE 19th International Conference on Control and Automation (ICCA)*, 2025.

[7] K. Guo, Y. Liu, J. Jia, **Z. Yang**, S. Zhou, X. Yu, and L. Guo, "Flying in harsh environments: Anti-rain quantification system and control strategy," *IEEE Transactions on Industrial Electronics*, 2025.

Talks

Presentation *IEEE* International Conference on Control and Automation (ICCA), Reykjavík, Iceland. — June, 2024

Internship

COMAC (Commercial Aircraft Corporation of China, Ltd.) Beijing Aircraft Technology Research Institute, Intern — July, 2022 - August, 2022

Honors and Awards

Oral Presentation, Top 1.8% The Thirteenth International Conference on Learning Representations (ICLR), Singapore. April, 2025

Best Student Paper Award *IEEE* International Conference on Control and Automation (ICCA), Reykjavík, Iceland. June, 2024

Funds

Second-Class Scholarship Beihang University, Beijing, China. 2024

Second-Class Scholarship Beihang University, Beijing, China. 2023

Skills

Coding: Python (skilled), Matlab/Simulink (skilled), C/C++ (familiar), HTML/CSS (basic)

Hardware: Skilled in UAV hardware-software co-design and system integration; developed fixed-wing and quadrotor platforms with embedded systems (Pixhawk, Jetson Nano, Intel NUC); experienced in rapid prototyping (3D printing, laser cutting), electronic assembly (soldering)...

Language: Mandarin Chinese (native), English (fluent, IELTS 7.5), Japanese (basic)

Misc.: Electric guitar, snowboarding, badminton, swimming...

Soft Skills.: Self-motivation, critical-thinking, multitasking...

Selected Projects

Robotics Related Projects

Observer-Enhanced Locomotion Policy for Quadruped Robots 2025
Beihang University, Unitree Robotics.

Lidar-Inertial Odometry and Mapping for Quadruped Robots 2025
Beihang University, Unitree Robotics.

Sucessive Convexification-based Planning and Disturbance-Aware Tracking Control of Quadrotor Swarms 2024
Beihang University, China Academy of Launch Vehicle Technology.

Design and Control Synthesis of a Flying-Wing UAV 2024
Beihang University.

Graph-based Multi-Robot Planning for Last-Mile Delivery 2023
Beihang University.

Material Science and Aerospace Engineering Related Projects

Finite Element Simulation of 3D-Printed Alumina Sintering Process 2023
Northwestern Polytechnical University (Undergraduate Thesis Project).

Optimal Transportation in a Rayleigh-Benard Turbulent Flow 2022
Northwestern Polytechnical University.

Deep CNN for Airfoil Inverse Design 2021
Northwestern Polytechnical University.

Potential Flow Solver for Lift Coefficient Prediction Under Flow Separation 2021-2022
Northwestern Polytechnical University (University Aeromodelling Team).

Design of 16kg-Cargo UAV with High Maneuverability 2021
Northwestern Polytechnical University (University Aeromodelling Team).